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Field of Testing : Chemical Testing and
 Microbiological Testing

Chemical Testing

Products	Specific Tests	Method/Reference
I. Foods		
.01 Bottled Water	Turbidity	Nephelometric Method, SMEWW
	Color (True)	Filtration, Visual Comparison, SMEWW 2120B
	pH	Glass Electrode Method, SMEWW
	Total Dissolved Solids	Gravimetric (dried at 108°C), SMEWW
	Conductivity	Laboratory Method, SMEWW
	Calcium	Flame AAS, Acid Digestion or Nitric Acid/Hydrochloric Acid Hotplate Digestion
	Magnesium	
	Sodium	
	Potassium	
	Aluminum	
	Chromium	
	Chloride	Ion Chromatography with Chemical Suppression of Eluent Conductivity Method, SMEWW 4110B
	Sulfate	
	Nitrate	
	Nitrite	
	Copper	Nitric Acid – Hydrochloric Acid Hotplate Digestion Method SMEWW 3030 F
	Iron	
	Manganese	

Cadmium	Inductively Coupled Plasma – Emission Spectroscopy Method SMEWW 3120
Lead	
Fluoride	Ion Selective Electrode Method, 4500F-C SMEWW
Surfactants (MBAS)	Methylene Blue Method, SMEWW
Arsenic	Manual Hydride Generation-AAS Method 3030F and 3111B, SMEWW
	Inductively Coupled Plasma - Emission Spectroscopy Method SMEWW 3120
Total Cyanide	With Distillation 4500CN-C Cyanide Selective Electrode Method, SMEWW
Free Cyanide	Without Distillation 4500CN-C Cyanide Selective Electrode Method, SMEWW
Mercury	Cold Vapor-AAS Method, SMEWW 3112 B
Phenol and Phenolic Substances (2-chlorophenol, 2,4-dichlorophenol and ,4,6-trichlorophenol)	Gas Chromatographic Mass Spectrophotometric Method, SMEWW 6420C / US EPA 8270
Benzene	Purge & Trap Capillary-Column Gas Chromatographic Mass Spectrophotometric Method, SMEWW 6200B / US EPA 8260
Pesticides and related substances (Organochlorines, Organophosphates and Polychlorinated Biphenyls)	Gas Chromatographic Mass Spectrophotometric Method, SMEWW 6410B / US EPA 8270

Legend to Reference Method/s

SMEWW – Standard Methods for the Examination of Water and Wastewater

***** *Nothing Follows* *****

Microbiological Testing

Products	Specific Tests	Method/Reference
I. Foods		
07.0 Cereals and cereal products		
07.1 Breakfast cereals	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast & Yeast-like Fungi	Pour Plate Method (BAM, Chapter 18)
	Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
07.2 Cereal/cereal grains	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.4 Soya flours, concentrates and isolates	Molds Count	Pour Plate Method (BAM, Chapter 18)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.5 Flour, corn meal, corn grits, semolina	Yeast & Yeast-like Fungi	Pour Plate Method (BAM, Chapter 18)
	Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
07.6 Dry Mixes (e.g. Pancake Mix, Purple Yam Mix)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
07.8 Soy protein	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Salmonella</i> Detection	Presence-Absence Method

		(BAM, Chapter 5)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
07.10 Pasta products and noodles Uncooked (wet and dry)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.11 Starch	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.12 Coconut Flour	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.13 Instant Noodles (Noodle Cake)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.14 Instant Noodles (mixed with seasoning)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)

	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
07.15 Rice based desserts (rice cakes), ready-to-eat e.g. kakanin, suman, layered, moulded, rice balls, mochi	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
08.0 Bakery Products		
08.1 Frozen bakery products (ready to eat) with low acid or high a_w fillings or toppings	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
08.2 Frozen bakery products (to be cooked) with low acid or high a_w fillings or toppings (e.g. meat pies and pizzas)	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
08.3 Frozen and refrigerated doughs	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
08.4 Baked goods (e.g. breads, cakes, pie crust, pastries and fried doughs)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast Count	Pour Plate Method (BAM, Chapter 18)
	Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
08.5 Coated or filled, dried shelf-stable biscuits	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Escherichia coli</i> Detection	MPN Method (BAM, Chapter 4)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)

10.0 Meat and meat products		
10.1 Dried animal products	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.2 Meat paste and paté (heat treated)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
	<i>Escherichia coli</i> Detection	MPN Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.3 Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	<i>Escherichia coli</i> Detection	MPN Method (BAM, Chapter 4)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.4 Packaged cooked cured/salted meat (ham, bacon)	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.6 Cooked poultry meat, frozen to be re-heated before eating (e.g. prepared frozen meals chicken burgers, chicken turkey rolls, chicken nuggets, other breaded poultry meat products)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.7 Cured/smoked poultry meat	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.8 Dehydrated poultry products	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)

10.10 Marinated Meat Products e.g. Marinated meat and meat preparations (tapa, sisig, etc.), Marinated Poultry, Dim Sum made from meat (siomai)	<i>Staphylococcus aureus</i> Count	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.11 Minced meat and meat preparations made from poultry meat intended to be eaten cooked	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.12 Minced meat and meat preparations made from species other than poultry intended to be eaten cooked	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
10.13 Foods cooked immediately prior to sale or consumption e.g. Takeaway food, burgers, kebabs, sausages, pizza, ready meals (cook/chill and cook/freeze) after regeneration	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	<i>Staphylococcus aureus</i> Count (coagulase +)	Spread Plating Method (CMMEF 5th ed.)
	<i>Salmonella</i> Detection	Presence-Absence Method (BAM, Chapter 5)
13.0 Beverages		
13.1 Non-alcoholic beverages (e.g. Ready to drink softdrinks, iced tea, energy drinks, jelly drinks)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
13.2 Frozen juice concentrate	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
13.4 Powdered beverages (e.g. iced tea, powdered juices/mixes)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
13.5 Fruit beverage products	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)
13.6 Bottled Water (Purified/Distilled Water)	Heterotrophic Plate Count	Pour Plate Method 9215B / SMEWW 23 rd ed.

	Coliform Count	Multiple Tube Fermentation Technique 9221B (SMEWW 23 rd ed.)
	<i>Escherichia coli</i> Detection	MPN Method (BAM, Chapter 4)
	Heterotrophic Plate Count	Pour Plate Method 9215B / SMEWW 23 rd ed.
13.7 Bottled Water (Mineral Water)	Coliform Count	Multiple Tube Fermentation Technique 9221B (SMEWW 23 rd ed.)
	<i>Escherichia coli</i> Detection	MPN Method (BAM, Chapter 4)
13.8 Chilled Young Coconut Water (Buko Juice)	Aerobic Plate Count	Pour Plate Method (BAM, Chapter 3)
	Yeast and Molds Count	Pour Plate Method (BAM, Chapter 18)
	Coliform Count	Solid Medium Method (BAM, Chapter 4)

Legend to Reference Method/s

BAM – Bacteriological Analytical Manual

CMMEF – Compendium of Methods for the Microbiological Examination of Foods

SMEWW – Standard Methods for the Examination of Water and Wastewater

***** *Nothing Follows* *****